

REQUIREMENTS SPECIFICATION DOCUMENT

E-Commerce Order Management Database System

Week1 | Database Management Systems

Field	Details
Student Name	LUCAS A
Register Number	5282510212
Department	Bsc(AI\ML)
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Bata Showroom Management System

1 Introduction / Overview

The Bata Showroom Management System is a software application developed to automate and manage the daily operations of a Bata footwear showroom. Bata is one of the leading footwear brands that offers a wide range of shoes, sandals, slippers, and accessories for men, women, and children. Managing a showroom manually can be challenging due to the large number of products, customers, sales transactions, and inventory records.

The proposed system helps showroom staff maintain product information, monitor stock levels, generate bills, manage customer details, and prepare sales reports efficiently. By computerizing these operations, the showroom can reduce errors, save time, improve productivity, and provide better customer service. The system also enables management to make informed business decisions through accurate reports and data analysis.

2. Scope

The scope of the Bata Showroom Management System includes the automation of various showroom activities related to product management, inventory control, billing, customer management, and reporting.

The system allows employees to add, update, and remove product details such as product name, size, color, category, quantity, and price. It tracks inventory levels and provides alerts when stock reaches a minimum level. Customer details and purchase history can be maintained for better service and future reference.

The billing module generates invoices automatically, reducing calculation errors. The system also produces daily, weekly, monthly, and yearly sales reports to help management evaluate business performance. The project is intended for use within Bata showrooms and can be expanded in the future to support multiple branches and online sales integration.

3. Problem Statement

Traditional manual methods of managing a Bata showroom involve maintaining records in registers, spreadsheets, or paper documents. These methods are time-consuming, prone to human errors, and difficult to maintain when the volume of products and customers increases.

Some common problems faced by manual systems include:

Difficulty in tracking inventory accurately.

Errors in billing and sales calculations.

Loss or damage of important records.

Time-consuming report generation.

Lack of real-time stock information.

Difficulty in analyzing sales performance.

Poor customer record management.

To overcome these challenges, a computerized showroom management system is required. The proposed system automates routine tasks, improves accuracy, enhances productivity, and provides better control over showroom operations.

4. Stakeholders

Stakeholders are individuals or groups who are directly or indirectly involved in the project.

Showroom Owner

The owner monitors business performance, reviews reports, and makes strategic decisions regarding inventory and sales.

Store Manager

The manager supervises daily operations, manages employees, checks stock levels, and ensures smooth functioning of the showroom.

Sales Staff

Sales personnel use the system to search products, generate bills, update sales records, and assist customers.

Customers

Customers benefit from accurate billing, faster service, and improved shopping experiences.

Inventory Manager

The inventory manager monitors stock availability, updates product quantities, and places orders for new stock when necessary.

Accounts Department

The accounts team uses sales and billing data for financial management, auditing, and tax-related activities.

5. Functional Requirements

Functional requirements describe what the system should do.

User Authentication

Users must log in using a valid username and password.

Different access levels should be provided for administrators and employees.

- Product Management
 - Add new products.
 - Update existing product details.
 - Delete discontinued products.
 - Search products by name, category, or size.
- Inventory Management
 - Track stock availability.
 - Update stock quantities automatically after sales.

- Generate low-stock alerts.
- Customer Management
- Store customer information.
- Maintain purchase history.
- Search customer records.
- Billing Management
- Generate invoices automatically.
- Calculate taxes and discounts.
- Print customer bills.
- Sales Management
- Record daily sales transactions.
- Monitor sales performance.
- Generate sales summaries.
- Report Generation
- Daily sales reports.
- Monthly sales reports.
- Inventory reports.
- Profit and loss reports.
- Administration
- Manage users and permissions.
- Monitor system activities.
- Maintain system settings.

6. Non-Functional Requirements

Non-functional requirements describe how the system should perform.

Performance

The system should process transactions quickly and generate reports without delays.

Security

Customer information, sales data, and financial records must be protected from unauthorized access.

Reliability

The system should operate consistently without failures during business hours.

Usability

The user interface should be simple and easy to understand for employees.

Availability

The system should be available whenever the showroom is open for business.

Scalability

The system should support future expansion such as multiple showroom branches.

Maintainability

The software should be easy to update and maintain.

Backup and Recovery

Regular backups should be taken to prevent data loss and support recovery in case of system failure.

7. System Modules

Login Module

Provides secure access through authentication and authorization.

Product Management Module

Maintains details of footwear products, brands, sizes, colors, prices, and categories.

Inventory Management Module

Tracks stock levels, updates inventory, and generates stock reports.

Customer Management Module

Stores customer information and purchase history.

Billing Module

Generates invoices, calculates totals, taxes, and discounts.

Sales Management Module

Records sales transactions and monitors business performance.

Report Generation Module

Produces various reports for management analysis.

Administration Module

Manages system users, permissions, and configurations.

8. Input Requirements

The system requires the following inputs:

- Product Information
- Product ID
- Product Name
- Brand Name
- Category

- Size
- Color
- Quantity
- Price
- Customer Information
- Customer ID
- Customer Name
- Phone Number
- Address
- Sales Information
- Bill Number
- Product Purchased
- Quantity Sold
- Payment Method
- Discount Details

Employee Information

- Employee ID
- Employee Name
- Username
- Password
- Inventory Information

Stock Quantity

- Supplier Details
- Reorder Level

9. Output Requirements

The system generates various outputs such as:

- Billing Output

- Customer invoice
- Purchase details
- Tax information
- Total amount
- Sales Reports
- Daily sales report
- Weekly sales report
- Monthly sales report
- Annual sales report
- Inventory Reports
- Current stock report
- Low-stock report
- Product availability report
- Customer Reports
- Customer purchase history
- Frequent customer details
- Management Reports
- Profit analysis
- Sales performance analysis
- Product-wise sales report

These reports help management evaluate business performance and make informed decisions.

10. Constraints

Budget Constraints

Development and maintenance costs may be limited by the showroom's budget.

Hardware Constraints

The system requires computers, printers, and networking equipment.

User Training

Employees need training to use the system effectively.

Data Accuracy

The system depends on accurate data entry by employees.

Internet Dependency

Certain features may require internet connectivity.

Security Risks

Unauthorized access and cyber threats must be prevented through proper security measures.

System Downtime

Hardware or software failures may temporarily affect showroom operations.

Future Expansion Limitations

Additional modifications may be required when expanding the system to multiple branches.